No Trading Strategy Can Win on Every Price Path: Computability, Randomness, and the Limits of Universal Trading

Karl Svozil [1]

[1] *Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8-10/136, A-1040 Vienna, Austria* [*]

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Can one algorithm make money on every computable price path? No. For any total computable trader, its code determines a total computable countermarket: simulate its next position and generate a bounded price move with the opposite sign. The resulting fixed path defeats that trader; no market need watch it in real time. A separate Halting-Problem reduction rules out universal inference of arbitrary computable market rules. Martin-Löf randomness and no-arbitrage provide different probabilistic and financial boundaries. Success therefore requires a restricted market class, distribution, benchmark, or informational advantage.

I. INTRODUCTION

Throughout this paper the elementary configuration has two sides: a *trader* and a *market*. At each date the trader converts the observed price history into a position; the market supplies the next price, and their interaction determines the gain. Depending on the model, "market" may mean a realized path, a program generating a path, or a probability law over paths. This organizing picture is not a claim that every financial model is literally a two-person zero-sum game.

A strategy is called *universal* here only if, after discounting by a specified numeraire, it produces a strictly positive terminal gain from zero initial capital on every admissible market trajectory. Zero cost, self-financing, admissibility, and discounted gain are essential: positive wealth obtained from a positive initial investment is not an arbitrage and need not outperform its financing benchmark. The universal claim has the quantifier order

$$\exists\, \sigma\ \forall\, \mathcal{E}: \qquad \sigma \text{ wins against the market } \mathcal{E}, \tag{1}$$

whereas the diagonal countermarket establishes

$$\forall\, \sigma\ \exists\, \mathcal{E}_\sigma: \qquad \sigma \text{ does not win against } \mathcal{E}_\sigma. \tag{2}$$

The second formula does not require a market that watches orders and reacts in real time. Fix a total computable trader. Its code can be incorporated into a new total program that recursively generates a price path: on the history generated so far, simulate the trader's next position and choose the next bounded price move with the opposite sign. Once the trader is fixed, this countermarket program and its entire infinite path are fixed as well; any finite prefix can be computed before deployment. A trader that never takes a position—that is, does not trade at all—earns zero rather than a strict loss, and zero already refutes a guarantee of strictly positive gain.

This is the trading version of the standard diagonal argument against universal computable prediction. Given a total next-bit predictor F, define the computable sequence $x_t = 1 - F(x_0, \ldots, x_{t-1})$. It disagrees with F at every step. For trading, complementing a bit becomes choosing a price move opposite to the position. The construction is tailored to the proposed universal rule: it proves $\forall\sigma\, \exists\mathcal{E}_\sigma$, not that one exceptional market defeats all conceivable traders.

Universal computation creates a second, distinct obstruction. If admissible market generators can simulate arbitrary Turing machines, a future price event can encode whether a computation halts. A program that decided every such event would decide the Halting Problem. Gold's identification-in-the-limit framework and later work on unsolvable inductive inference place this explicit reduction in a broader learning-theoretic setting [1–5].

At the passive pole, a binary price record may be Martin-Löf random relative to a specified computable fair measure. It then passes every effective null test and permits no lower-semicomputable nonnegative fair-game capital process to become unbounded. This does not exclude isolated correct forecasts or finite lucky profits. No-arbitrage supplies yet another boundary, formulated almost surely under a pricing measure rather than through an adversarial countermarket or an effective null test.

The market may also cease to be exogenous when the trader is large, crowded, or publicly known: orders consume liquidity, move quotes, reveal information, and attract imitation or opposition. Any defensible claim of repeatable success must therefore identify a restriction, asymmetry, benchmark, or distributional assumption absent from the unrestricted market class. A trend follower posits persistence, a contrarian posits mean reversion, an option seller relies on a risk premium, and an informed trader relies on information not yet fully reflected in prices.

The paper's principal contribution is this taxonomy and synthesis, not a claim that the diagonal countermarket is an equilibrium model or a newly discovered impossibility theorem. The mathematical study of unpredictable prices runs from Bachelier through Samuelson to the Efficient Market Hypothesis [6–8]; the present purpose is to keep separate several modern meanings of "no univer-

[*] karl.svozil@tuwien.ac.at; http://tph.tuwien.ac.at/~svozil

TABLE I. Distinct claims organized by the trader–market configuration. They are complementary, not interchangeable.

Mechanism	**Market side**	**Formal scope**	**Conclusion**
Diagonal countermarket	Fixed total program constructed from the fixed trader's code	Pathwise worst case: $\forall\sigma\,\exists P_\sigma$	No computable trader wins on every computable price path
Halting reduction	Total program or generator, under an explicit promise	Individual encoded computation over an unbounded horizon	No general computable profit or market-event decider
Algorithmic randomness	Passive binary record under a specified computable fair measure	Each Martin-Löf-random path; such paths have measure one	No effective fair-game capital process becomes unbounded
No-arbitrage	Stochastic price process admitting a coherent pricing measure	Almost-sure payoff conditions under an ELMM	No admissible zero-cost classical arbitrage
Cover universality	Passive bounded sequence and a stated benchmark class	Pathwise, but benchmark-relative and asymptotic	Vanishing per-period regret, not guaranteed positive profit

sal strategy." Section II gives the diagonal countermarket, two Halting reductions, and the Martin-Löf, computable, and Schnorr randomness hierarchy. Sections III and IV give the financial and combinatorial boundaries. Section V treats time reversal only as a practical robustness diagnostic. The remaining sections compress the social-choice analogy, analyze the Wheel, distinguish Cover's benchmark-relative result, and state the conditional scope of FAPP strategies.

II. THE TRADER AND THE MARKET: COMPUTABILITY AND RANDOMNESS

We now formalize the trader–market pair. The trader is a program M_σ mapping each observed finite price history to a position; the market side supplies the price history. Three specifications lead to three different boundaries: a fixed computable countermarket tailored to a trader defeats its claim of pathwise universality, a computationally universal class of market generators makes unrestricted future-event inference undecidable, and a passive algorithmically random record defeats effective fair betting.

A. A Computable Countermarket for Every Trader

Let M_σ be a *total* computable map from every finite computable price history H_t to a rational position w_t. Totality models the practical requirement that a deployed algorithm return an action before a deadline. Let $\mathcal{P}^+_{\mathrm{comp}}$ denote the positive rational-valued computable price paths.

Theorem 1 (Computable Diagonal Countermarket). *For every total deterministic computable strategy M_σ, there exists a fixed path $P^{M_\sigma} \in \mathcal{P}^+_{\mathrm{comp}}$ on which its self-financing gain*

$$G_T = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t)$$

is non-positive at every finite horizon T. It is strictly negative after the strategy has taken a nonzero position at least once.

Proof. Choose rational $P_0 > 0$ and $0 < \epsilon < 1$. Recursively compute $w_t = M_\sigma(P_0, \dots, P_t)$ and set

$$P_{t+1} = \begin{cases} P_t(1-\epsilon), & w_t > 0, \\ P_t(1+\epsilon), & w_t < 0, \\ P_t, & w_t = 0. \end{cases} \tag{3}$$

Because M_σ is total and rational-valued, this recursion is a total program for one fixed infinite price path. Its prices remain positive and computable. In each period, $w_t(P_{t+1} - P_t) \le 0$, with strict inequality when $w_t \ne 0$. Summing proves the claim at every finite horizon. □

Remark 1 (Logical and Economic Scope of the Countermarket). Theorem 1 uses the trader's *code*, not a physical market's observation of a deployed order. Once M_σ is fixed, the countermarket program is fixed and can generate any desired finite prefix offline. It is therefore a lawful, deterministic, computable counterexample, although not necessarily an efficient one computationally. If the market side is regarded as an opponent whose objective is to prevent this trader's gain, the rule is rational relative to that objective. It is not thereby shown to be an economic equilibrium or an arbitrage-free price process, and another trader may profit on it. Such a trader has, in turn, its own computable countermarket.

This is a direct strategy-relative diagonal construction. The proposed universal trader is used to compute an object outside its success set: long is paired with down, short with up, and neutrality with no price change. Just as the complement sequence defeats a purported predictor of every computable sequence, P^{M_σ} defeats the purported trader on every horizon. Enumeration of all programs is unnecessary because the universal claim itself supplies the program to be diagonalized against. The quantifiers are $\forall M_\sigma\, \exists P^{M_\sigma}$. There need not be one path on which every strategy loses; for example, permanently long and permanently short positions cannot both lose on the same one-step price move.

B. Universal Computation and the Halting Barrier

The preceding diagonal construction already disproves a computable trader that wins on every computable path; it does not invoke the Halting Problem. A separate question asks whether one general program can inspect arbitrary encoded traders or market generators and correctly decide their unbounded future behavior. This certification and rule-inference problem is undecidable.

Theorem 2 (Undecidability of Eventual Profit)**.** *Fix the positive computable price path $P_t = t+1$, $t \in \mathbb{N}$. There is no algorithm which, under the promise that its input code computes a total trading program M_σ, correctly decides for every such code whether*

$$\exists T \geq 1 : \qquad G_T(M_\sigma; P) = \sum_{t=0}^{T-1} w_t(P_{t+1} - P_t) > 0.$$

Proof. Assume such a profit certifier exists. Given an arbitrary Turing machine A and input x, construct a trading program $M_{A,x}$ as follows. At date t, simulate $A(x)$ for $t+1$ steps. If it has halted within those steps, set $w_t = 1$; otherwise set $w_t = 0$. Every date requires only a finite simulation, so $M_{A,x}$ is total. Along $P_t = t+1$, every long position earns one unit in the next period. Hence $G_T > 0$ for some T if and only if $A(x)$ eventually halts. A certifier for eventual trading profit would therefore decide the Halting Problem, which is impossible [9]. □

The reduction can also be placed directly on the market side, which is the form most closely connected to general rule inference.

Theorem 3 (Undecidability of a Market Event)**.** *There is no algorithm which, under the promise that its input code computes a total positive-price generator G, correctly decides for every such code whether its generated path P^G ever satisfies $P^G_t > P^G_0$.*

Proof. Given a Turing machine A and input x, define the total generator by $P_0^{A,x} = 1$ and, for $t \geq 1$,

$$P_t^{A,x} = 1 + \mathbf{1}\{A(x) \text{ halts within } t \text{ steps}\}, \qquad t \geq 1. \tag{4}$$

Each quoted price requires only a finite t-step simulation, so the generator is total and computable. Its initial price is one, and it ever quotes a higher price if and only if $A(x)$ halts. A universal decider for the stated market event would therefore decide the Halting Problem. □

Theorem 3 makes the assumption of universal computation precise: the admissible language of market generators must be rich enough to emulate an arbitrary machine and expose the result through an observable price event. Under that condition, no computable rule-inference procedure can uniformly settle all future events of all such markets. This is a concrete market version of the limits studied in identification in the limit and in unsolvable inductive inference [1–5, 10]. It does not deny that restricted parametric, stochastic, or otherwise structured model classes can be learned.

The three computational claims are related but not identical. Theorem 1 has quantifiers $\forall M_\sigma\, \exists P^{M_\sigma}$ and constructs a fixed computable countermarket from the code of a fixed trader; this is direct diagonalization, not a reduction from halting. Theorem 2 embeds an arbitrary computation in the trader while fixing a very simple rising market. Theorem 3 embeds it in the market and rules out a general future-event inference procedure. Thus diagonalization refutes the universal strategy, while the Halting reductions refute universal certification and rule inference. For a specified finite horizon, by contrast, total programs can simply be run and their realized gains and prices computed.

Corollary 4 (Maximin Value)**.** *At every finite horizon T, if the neutral strategy is allowed, then*

$$\sup_{\text{total computable } M_\sigma} \inf_{P \in \mathcal{P}^+_{\text{comp}}} G_T(M_\sigma, P) = 0.$$

Proof. The neutral strategy guarantees zero, while Theorem 1 gives every strategy a computable countermarket with gain at most zero. □

Proposition 5 (Private Randomization Does Not Create a Guarantee)**.** *Suppose a randomized trader's current coin toss is hidden from the environment. There still exists a passive computable distribution of next returns under which every integrable strategy has conditional expected gain zero.*

Proof. Let the next price increment be $+\epsilon P_t$ or $-\epsilon P_t$ with equal probability, independently of the trader's private draw. Conditional on the current information and position, $\mathbb{E}[w_t(P_{t+1} - P_t) \mid \mathcal{F}_t] = 0$. Summing gives expected gain zero. □

Private randomization prevents a deterministic countermarket based only on the public history from mechanically opposing the unseen current draw, but it does not yield strictly positive expected gain in every environment.

C. The Optional Online-Game Reading

The recursive construction in Theorem 1 admits an equivalent finite perfect-information game reading: the trader announces w_t, the environment chooses the next return, and the terminal payoff is G_T. Its value is zero by Corollary 4. This sequential presentation is optional, however. For a fixed deterministic trader, running the already fixed countermarket program produces exactly the same path; actual observation of a deployed trader is not a premise. For infinite perfect-information games, Borel determinacy guarantees that one player has a pure winning strategy when the winning set is Borel [11], but the theorem does not identify the winning player without analysis of the payoff set. Nor does it cover a trader's hidden random bits: that is an imperfect-information game whose relevant conclusion here is the mixed value zero in Proposition 5.

D. A Passive Market: Martin-Löf Randomness

Unlike the tailored countermarket, the path considered here is not constructed from the trader's code. Suppose an effective coding of price directions is Martin-Löf random relative to a specified computable fair measure. Here "random" means algorithmically random, not merely sensitive dependence in deterministic chaos. Such a sequence may still be guessed correctly on many individual dates. The exact claim is that it passes every effective null test, or equivalently that no lower-semicomputable nonnegative fair-game capital process becomes unbounded on it.

In the fair binary market, a nonnegative capital process $K\colon \{0,1\}^{<\mathbb{N}} \to \mathbb{R}_{\geq 0}$ is a martingale when

$$K(s) = \frac{K(s0) + K(s1)}{2}. \tag{5}$$

Theorem 6 (Ville's Inequality)**.** *If $K_0 = 1$ is a nonnegative martingale under the fair-coin measure, then for every $\lambda > 0$,*

$$\mathbb{P}\left(\sup_{n\geq 0} K_n \geq \lambda\right) \leq \frac{1}{\lambda}. \tag{6}$$

Ville's inequality follows by stopping the nonnegative capital process when it first crosses λ and applying optional stopping, followed by a limit [12]. In particular, the set on which a fixed nonnegative martingale becomes unbounded has probability zero.

To state the effective refinement correctly, let λ denote fair-coin measure on Cantor space $\{0,1\}^{\mathbb{N}}$. A Martin-Löf test is a uniformly computably enumerable sequence of open sets $(U_k)_{k\geq 1}$ satisfying $\lambda(U_k) \leq 2^{-k}$. A sequence is *Martin-Löf random* if it avoids $\bigcap_k U_k$ for every such test [13]. A sequence is *computably random* if no total computable nonnegative martingale succeeds on it by becoming unbounded. A *Schnorr-random* sequence passes every Martin-Löf test whose component measures are uniformly computable. These are distinct notions, with strict implications

$$\begin{aligned}\text{Martin-Löf random} &\implies \text{computably random} \\ &\implies \text{Schnorr random.}\end{aligned} \tag{7}$$

Neither converse holds [14–16].

Theorem 7 (Effective Martingale Barrier)**.** *A binary sequence x is Martin-Löf random if and only if no lower-semicomputable nonnegative supermartingale succeeds on x by unbounded capital. Consequently, the success set of each such betting strategy is effectively λ-null, and the set of paths on which all of them remain bounded has fair-coin measure one.*

Proof. Normalize a lower-semicomputable nonnegative supermartingale d by $d(\varnothing) \leq 1$, and let

$$U_k = \{x : \exists n,\ d(x{\upharpoonright}n) > 2^k\}.$$

Lower semicomputability makes U_k computably enumerable and open, while Ville's inequality gives $\lambda(U_k) \leq 2^{-k}$. Hence (U_k) is a Martin-Löf test and no Martin-Löf-random sequence permits d to become unbounded. Conversely, every Martin-Löf test can be converted into a lower-semicomputable supermartingale that succeeds on its intersection; this is the martingale characterization of Martin-Löf randomness [15, 17]. ☐

The same construction extends from λ to any computable probability measure μ with the supermartingale condition and cylinder probabilities defined relative to μ. This is the effective analogue most directly applicable to a computable martingale measure.

Theorem 7 contains the proposed "typical path" claim in its correct form. Under a computable fair measure, a measure-one class of paths simultaneously blocks unbounded success by every effective nonnegative fair-game capital process. It does *not* say that every such path gives every strategy a finite loss, nor does it apply unchanged to strategies with unrestricted borrowing or to non-martingale physical measures. There is also a prediction corollary under explicit assumptions. Consider binary outcomes under the fair-coin reference measure, a total computable next-bit predictor, no costs or asymmetric payoffs, and a fixed positive asymptotic advantage $\liminf_{T\to\infty} A_T > 1/2$. Betting a sufficiently small rational fraction $\delta \in (0,1)$ of current capital on each prediction multiplies capital by $1+\delta$ after a correct prediction and by $1-\delta$ after an incorrect one. For sufficiently small δ, the assumed accuracy advantage gives positive asymptotic logarithmic growth and therefore an unbounded computable martingale. Thus no such sustained directional advantage is possible on a Martin-Löf-random fair-coin path. This is not a direct claim about trading profit with costs, unequal return magnitudes, or financing constraints; finite streaks and isolated correct forecasts remain possible.

Theorem 1 is different: its fixed computable market is constructed from one deterministic trader and gives that trader non-positive gain at every finite horizon. A Martin-Löf-random market is not tailored to one trader and simultaneously passes all effective tests. The former is a strategy-relative diagonal counterexample; the latter is a typical-path statement relative to a specified computable measure. Game-theoretic probability develops this pathwise betting viewpoint systematically [18].

E. What Rice's Theorem Does and Does Not Say

Rice's theorem states that every nontrivial extensional property of arbitrary partial computable functions is undecidable [19]. It therefore limits fully general program certification; even deciding whether arbitrary code is total is impossible in general [9]. It does *not* imply that the loss-producing inputs of a fixed total finite-horizon strategy are undecidable. On a specified finite computable path, such a program can be simulated and its gain calculated. No probability claim about market failure sets follows from Rice's theorem alone. Theorem 2 supplies the exact statement needed here by an explicit Halting-Problem reduction: it is the unbounded question of *eventual* profit, not the evaluation of a completed finite trade record, that is undecidable in general. Theorems 2 and 3 are promise-domain results: their hypothetical deciders are required to be correct on codes promised to compute total objects, and the reductions themselves always produce codes satisfying that promise.

III. THE NO-ARBITRAGE FOUNDATION

Here the market is no longer an opponent choosing a response after seeing the trader's action. Instead, financial admissibility and no-arbitrage delimit which trader–market pairs are coherent.

A. Discounted Gains and Admissibility

Let the market be defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{P})$. Let $B_t > 0$ be a numeraire and S_t the vector of traded asset prices. Write $\overline{S}_t = S_t/B_t$ for discounted prices. A predictable, $\overline{S}$-integrable strategy H has discounted self-financing gain

$$G_t = (H \cdot \overline{S})_t, \qquad G_0 = 0. \tag{8}$$

It is *admissible* if there is a finite constant a such that

$$G_t \geq -a \quad \text{for all } t \in [0,T], \quad \mathbb{P}\text{-a.s.} \tag{9}$$

This uniform lower bound rules out doubling systems financed by unbounded credit.

Theorem 8 (Universality Implies Classical Arbitrage)**.** *If an admissible self-financing strategy H with zero initial capital satisfies*

$$G_T \geq 0 \quad \mathbb{P}\text{-}a.s., \qquad \mathbb{P}(G_T > 0) > 0,$$

then H is a classical arbitrage. In particular, a pointwise universal strategy is a classical arbitrage whenever its gain is defined on an admissible path class carrying $\mathbb{P}$.

Proof. The displayed payoff conditions, together with zero initial cost, self-financing, and admissibility, are the standard classical-arbitrage conditions. Pointwise strict positivity on every admissible path implies them, with $\mathbb{P}(G_T > 0) = 1$. ☐

Remark 2 (Terminology and Strength). The pathwise condition is sometimes called a sure or strong arbitrage, but it is not synonymous with *arbitrage of the first kind*. The latter is the condition dual to No Unbounded Profit with Bounded Risk (NUPBR/NA1) and is a distinct, weaker viability concept [20]. Classical no-arbitrage alone already excludes the universal strategy in Theorem 8; NFLVR is invoked below because it supplies the standard martingale-measure formulation.

B. NA, NUPBR, and Benchmark-Relative Gains

The relevant no-arbitrage notions do not form the simple chain sometimes suggested in informal discussions. In the standard semimartingale setting, NFLVR combines classical NA with NUPBR, whereas NA and NUPBR are in general logically distinct. NUPBR is equivalent, under standard portfolio hypotheses, to the existence of a numeraire portfolio whose positive relative-wealth processes are supermartingales [20, 21]. It may hold even when the stronger NFLVR condition and an ELMM fail.

This distinction prevents two overstatements. First, Theorem 8 uses only classical NA; NUPBR alone is not silently substituted for it. If a zero-cost gain were nonnegative at all intermediate times and positive at terminal time, scaling would violate NUPBR, but terminal pathwise positivity together with only a strategy-specific lower bound does not by itself supply that scaling argument. Second, benchmark and stochastic-portfolio approaches can generate gains *relative* to a numeraire or market portfolio without producing zero-cost positive gain on every path. Relative arbitrage in diverse equity markets and real-world benchmark pricing therefore do not contradict the theorem [22, 23].

C. The Martingale Constraint

For a locally bounded semimartingale price process, the Delbaen–Schachermayer form of the First Fundamental Theorem states that No Free Lunch with Vanishing

Risk (NFLVR) is equivalent to the existence of an equivalent local martingale measure (ELMM) $\mathbb{Q} \approx \mathbb{P}$ under which $\overline{S}$ is a local martingale [24, 25]. For general unbounded semimartingales the corresponding formulation uses an equivalent σ-martingale measure [26]; the corollary below retains the locally bounded setting.

Corollary 9 (No Universal Zero-Cost Strategy)**.** *If an ELMM $\mathbb{Q}$ exists, no admissible self-financing strategy with zero initial capital can satisfy $G_T \geq 0$, $\mathbb{P}$-almost surely, and $\mathbb{P}(G_T > 0) > 0$. Consequently, no pointwise universal strategy exists.*

Proof. Under $\mathbb{Q}$, the admissible stochastic integral $G = H \cdot \overline{S}$ is a local martingale bounded from below and therefore a supermartingale. Hence $\mathbb{E}_{\mathbb{Q}}[G_T] \leq G_0 = 0$. If the displayed arbitrage conditions held, equivalence would give $G_T \geq 0$, $\mathbb{Q}$-almost surely, with $\mathbb{Q}(G_T > 0) > 0$. Consequently $\mathbb{E}_{\mathbb{Q}}[G_T] > 0$, a contradiction. $\square$

The ELMM conclusion concerns discounted gains under the pricing measure $\mathbb{Q}$. It does not say that every risky strategy has non-positive expected return under the physical measure $\mathbb{P}$; positive physical expected returns may compensate risk.

D. Win Rate Is Not Expected Value

The probability of closing a trade at a profit can greatly exceed $1/2$ even in a fair market. Optional stopping gives the precise distinction.

Proposition 10 (Asymmetric Barriers in a Fair Random Walk)**.** *Let X_n be a simple symmetric random walk with $X_0 = 0$, and for positive integers a, b let*

$$\tau = \inf\{n \geq 0 : X_n = a \text{ or } X_n = -b\}.$$

Then

$$\mathbb{P}(X_\tau = a) = \frac{b}{a+b}, \qquad \mathbb{E}[X_\tau] = 0. \tag{10}$$

Proof. The stopped process $X_{n \wedge \tau}$ is a bounded martingale, so Doob's optional-stopping theorem applies [27]. If $p = \mathbb{P}(X_\tau = a)$, then $0 = \mathbb{E}[X_\tau] = ap - b(1-p)$, which gives $p = b/(a+b)$. $\square$

With a take-profit of $a = 1$ and a stop-loss of $b = 9$, the win probability is 0.9, but the expected payoff is zero: nine frequent unit gains are offset by one nine-unit loss in expectation. Thus prediction accuracy, trade win rate, and expected discounted profit are three different quantities.

IV. THE COMBINATORIAL PERSPECTIVE

Here the opposing side is an ensemble of market records rather than an adaptive market. The optimization NFL theorem and the elementary prediction result needed for market-direction claims should be distinguished.

Theorem 11 (Optimization NFL, Wolpert–Macready)**.** *For finite search and cost spaces, when performance is averaged uniformly over all objective functions, any two non-revisiting black-box search algorithms have identical averaged performance [28].*

The theorem concerns oracle-based optimization over function spaces. It does not, by itself, imply a statement about self-financing gains on stochastic price paths. A sharpening makes its structural assumption explicit: for a uniform distribution on a finite function class, the NFL equality for all non-revisiting algorithms holds precisely when the class is closed under permutations of the search domain [29]. Financial path classes are generally not permutation-closed: temporal order, continuity, financing, and no-arbitrage restrictions distinguish one permutation from another. Thus a financial edge, when present, resides in exactly such symmetry-breaking structure and the inductive bias used to model it.

For binary directional prediction the relevant result is simpler and can be derived directly.

Proposition 12 (Uniform Binary Prediction)**.** *Let $X_1, \ldots, X_T$ be uniformly distributed on $\{0,1\}^T$. A causal deterministic predictor chooses $\widehat{X}_t = f_t(X_1, \ldots, X_{t-1})$. Its mean accuracy*

$$A_T = \frac{1}{T} \sum_{t=1}^{T} \mathbf{1}\{\widehat{X}_t = X_t\}$$

satisfies $\mathbb{E}[A_T] = 1/2$. The same holds for a randomized predictor whose private randomization is independent of the next bit.

Proof. Conditional on every history $X_1, \ldots, X_{t-1}$, the next bit X_t is an independent fair bit. Therefore $\mathbb{P}(\widehat{X}_t = X_t \mid X_1, \ldots, X_{t-1}) = 1/2$. Linearity of expectation gives the result. Conditioning additionally on the predictor's private randomization proves the randomized case. $\square$

Consequently, no predictor can achieve expected directional accuracy greater than $1/2$ under the uniform distribution over all binary sequences; in particular, none can exceed $1/2$ on every sequence. This is an NFL symmetry statement, not a model of actual markets. Financial sign sequences are not uniform in general, and a directional forecast does not determine trading profit because payoff sizes, financing, and stopping rules matter (Proposition 10).

The defensible practical conclusion is conditional: a claim of repeatable success must identify non-uniform structure in the physical measure $\mathbb{P}$, such as a trend, mean reversion, a risk premium, information, or an institutional constraint that breaks the NFL symmetry. The hypothesized edge concerns a restricted environment, not universal algorithmic superiority.

V. A PRACTICAL ROBUSTNESS DIAGNOSTIC: TIME REVERSAL

This section supplies a falsification tool, not a further impossibility theorem on the level of the Halting reductions or no-arbitrage. The trader–market viewpoint suggests holding the trader fixed while transforming the market record. Among computably specified stress transformations, time reversal is perhaps the simplest, particularly for detecting dependence on directional persistence.

Let $\mathcal{M}$ be a space of strictly positive paths $P\colon [0,T] \to \mathbb{R}^n_{>0}$. The time reversal of P is

$$\widetilde{P}(t) = P(T-t), \qquad t \in [0,T]. \tag{11}$$

For a causal strategy σ, let $\Pi_T[\sigma; P]$ denote its terminal discounted gain from zero initial capital when evaluated on P.

Definition 1 (Pathwise Universal Strategy)**.** A causal strategy σ is *pathwise universal* on $\mathcal{M}$ if it is self-financing and admissible and satisfies $\Pi_T[\sigma; P] > 0$ for every $P \in \mathcal{M}$.

Criterion 1 (Reversal Falsification Test)**.** Suppose $\mathcal{M}$ is closed under time reversal. If there exists $P \in \mathcal{M}$ such that

$$\min\{\Pi_T[\sigma; P], \Pi_T[\sigma; \widetilde{P}]\} \le 0, \tag{12}$$

then σ is not pathwise universal on $\mathcal{M}$.

Proof. Both P and $\widetilde{P}$ belong to $\mathcal{M}$. A non-positive gain on either member of the pair contradicts Definition 1. □

Criterion 1 is deliberately a falsification device, not an independent impossibility theorem. Reversing a profitable path does not in general negate the profit of a causal strategy, because the strategy can take different positions on the reversed history. The value of the test is that it generates a paired stress scenario whenever the modeled path class is reversal-closed.

Remark 3 (Computable Stress Tests Beyond Reversal). Time reversal is only the first member of a much larger battery. Depending on the admissible market class, a computable probe can reverse return signs, splice trend and anti-trend regimes, insert jumps or observation delays, rescale volatility, or add financing and transaction costs. Because programs have finite descriptions, computably specified probes form an at most countable family, and countably infinite test batteries can be constructed. Each transformed record must still belong to the market class against which the strategy's claim is made.

The word "test" is used here in the empirical sense of stress and falsification. A Martin-Löf test is a different technical object: a uniformly computably enumerable sequence of open sets with effective measure bounds. The shared idea is to confront an algorithm with many effective challenges; time reversal by itself neither defines a null set nor certifies randomness. Passing it establishes limited robustness, never universality.

Remark 4 (Filtrations and Time Reversal). In stochastic finance, the set of admissible trajectories is not automatically closed under time reversal. The process $P(T-t)$ is generally not adapted to the original forward filtration, and its semimartingale property under a reversed filtration requires additional hypotheses; time reversal of diffusions is a separate theorem, not a formal substitution $t \mapsto T-t$ [30].

A simple Black–Scholes calculation illustrates both the useful case and its limit. If

$$X_t = \log(S_t/S_0) = mt + \sigma W_t,$$

then the reversed increment $Y_t = X_{T-t} - X_T$, relative to its reversed filtration, has the law of a Brownian motion with drift $-m$ and volatility σ. Thus a log-price model class containing both drift signs is closed in law under reversed increments. Under the physical measure, $m = \mu - \sigma^2/2$; under a risk-neutral measure the discounted stock is a martingale, but its logarithm still has Itô drift $-\sigma^2/2$. Consequently, neither an ELMM nor a driftless arithmetic Brownian special case establishes detailed balance for a general financial model. The stress test remains pathwise; the rigorous probabilistic constraint is given next.

VI. A BRIEF SOCIAL-CHOICE ANALOGY

Arrow's Impossibility Theorem concerns social welfare functions on at least three alternatives. Under unrestricted preference profiles, collective rationality, Pareto unanimity, and independence of irrelevant alternatives, the aggregation rule must be dictatorial [31]. Its objects and assumptions differ from those of trading, and neither theorem proves the other. The limited comparison is one of domain: Arrow quantifies over all admissible preference profiles, while pathwise universality quantifies over all admissible price paths. In both cases an unrestricted domain makes a natural collection of requirements jointly untenable.

Yanofsky shows how many self-referential arguments can be organized using fixed points and fixed-point-free transformations [32]. The response in Eq. (3) anti-aligns returns with every nonzero position, but it admits the neutral fixed point $w_t = 0$. That fixed point yields zero gain rather than strict profit. The relevant obstruction is therefore the absence of a *strictly profitable* fixed response, not the absence of every fixed point.

VII. CASE STUDY: THE WHEEL OPTIONS STRATEGY

The Wheel sells a cash-secured put and, after assignment, sells a covered call against the acquired stock. Its economic exposure follows directly from put–call parity. Ignoring dividends and financing for the terminal-payoff identity,

$$S_T - (S_T - K)^+ = K - (K - S_T)^+. \tag{13}$$

The left-hand side is a covered-call payoff; the right-hand side is cash paying K plus a short put. After the corresponding time-zero costs are matched, a covered call and a cash-secured put are equivalent European payoffs. Rolling the Wheel therefore repeatedly maintains a put-like, short-downside-convexity exposure rather than creating directionless income. Equivalently,

$$\Phi_K(S_T) = S_T - (S_T - K)^+ = \begin{cases} S_T, & S_T < K, \\ K, & S_T \geq K, \end{cases} \tag{14}$$

which makes the one-for-one downside and capped upside explicit.

Option-selling returns may nevertheless have a positive physical mean. The difference between risk-neutral and physical expectations of volatility is a variance risk premium; option-market evidence documents compensation for bearing volatility and jump risk [33, 34]. This is an empirical, distribution-dependent explanation for some Wheel income, not a pathwise identity: the sign and magnitude depend on prices, strikes, transaction costs, and the physical law.

A. Failure I: Catastrophic Drop

If the underlying crashes after put assignment, the short call liability decreases, which benefits the call writer, but the loss on the stock dominates and the combined covered-call position can suffer a large net loss. Equation (13) makes this explicit: below K, the payoff falls one-for-one with S_T. A reversed rally/crash pair can be used as the stress test in Criterion 1, but time reversal alone is not the proof; the payoff identity is.

B. Failure II: Persistent Decline

Repeated option premiums offset part of a persistent decline but do not remove the renewed short-put exposure. The loss is path-dependent rather than generally linear: strikes, maturities, volatility, assignment dates, and position sizing determine the sequence of gains.

C. Failure III: Violent Breakout and Benchmark Underperformance

In a large rally, shares may be called away at the call strike. The Wheel can still realize a positive absolute profit, so opportunity cost alone is not a counterexample to strict positive P&L. It is instead a failure relative to a buy-and-hold benchmark: Eq. (13) caps the terminal payoff at K. No-arbitrage constrains the price of this payoff under $\mathbb{Q}$; it does not say that its physical expected return must be zero.

D. Failure IV: Absorbing Ruin Under Capital Constraints

A fully cash-secured, unlevered cycle has bounded loss, but an underlying that falls to zero can consume almost all committed capital net of premium. Leverage makes the absorbing nature of ruin explicit. If wealth evolves as $W_{t+1} = W_t(1+fR_{t+1})$ and an admissible return satisfies $1 + fR_{t+1} = 0$, then $W_{t+1} = 0$; a self-financing strategy cannot recover without an external capital injection. This is a wealth-dynamics fact, not a consequence of undecidability.

Table II is a payoff-based classification, not a claim that each trajectory has positive probability under every continuous price model. Corollary 9 excludes strict positive discounted gain on all paths, but probabilities of particular failure classes require a specified physical measure.

Repeated option income also raises a time-versus-ensemble distinction. For multiplicative wealth $W_T = W_0 \prod_{t=1}^{T}(1 + fR_t)$, the realized per-period growth is

$$\frac{1}{T} \log \frac{W_T}{W_0} = \frac{1}{T} \sum_{t=1}^{T} \log(1 + fR_t),$$

not the logarithm of an ensemble-average payoff. Frequent small gains can coexist with poor long-run growth when rare losses are large, a point emphasized in discussions of non-ergodic economic growth [35]. This does not replace expected-utility analysis; it identifies which long-run observable is being optimized.

VIII. UNIVERSAL PORTFOLIOS AND DISTANCE FROM UNIVERSALITY

The word "universal" also appears in a celebrated positive theorem. For price-relative vectors $x_t \in \mathbb{R}^m_{>0}$, a constant-rebalanced portfolio b has wealth

$$S_T(b) = \prod_{t=1}^{T} b^\top x_t.$$

Cover's universal portfolio is nonanticipating and, under the conditions of his theorem, has the same asymptotic exponential growth rate as the best constant-rebalanced portfolio chosen in hindsight on every individual bounded sequence [36]. In regret notation,

$$\frac{1}{T} \left[\log \max_b S_T(b) - \log S_T^{\mathrm{UP}} \right] \longrightarrow 0. \tag{15}$$

TABLE II. Wheel failure modes and the mathematical mechanism illustrating each one.

Failure	**Trajectory Class**	**Relevant Result**	**Mechanism**
I: Crash	Large downside move	Eq. (13)	Put-like downside
II: Bleed	Persistent negative drift	Repeated payoff exposure	Premiums do not insure the stock
III: Breakout	Large upside move	Eq. (13)	Capped benchmark return
IV: Ruin	Capital exhaustion	Wealth recursion	Zero wealth is absorbing

Equation (15) is a relative guarantee, not an absolute-profit guarantee. If every constant-rebalanced portfolio loses money on a path, the universal portfolio may lose as well while retaining vanishing per-period regret. Cover's result therefore does not contradict Corollary 9; it illustrates the strongest useful meaning of "universal" once the benchmark class is specified.

a. A Hierarchy of Attainable Claims The useful alternatives to pathwise universality are not weaker points on one single order; they change the benchmark, horizon, or probability quantifier. Table III records the distinctions.

Superhedging duality gives another quantitative notion: once a contingent claim and an admissible model class are fixed, its superhedging price is the least initial capital needed for a pathwise guarantee. Requiring zero initial capital while demanding a strictly positive payoff is precisely the degenerate, arbitrage-producing endpoint ruled out above.

IX. FAPP STRATEGIES AND THEIR HONEST SCOPE

While pathwise universal profit is precluded, strategies may be profitable *for all practical purposes* (FAPP, following Bell [37]) relative to a physical measure $\mathbb{P}$, a benchmark, and a finite horizon. Market making may earn spreads when adverse selection and inventory risk are controlled, and Kelly investing maximizes expected logarithmic growth when the relevant distribution is sufficiently well specified [38]. Neither is an unconditional profit guarantee.

Here "insight" means an explicit restriction rather than a claim about a trader's state of mind. A strategy specifies a market class and a mechanism expected to persist in it, and the resulting claim is falsifiable outside that class. A lucky finite gain needs no such account; a defensible claim of repeatable edge, positive conditional expectation, or sustained benchmark outperformance must state one.

Economic equilibrium also leaves room for conditional edges. Grossman and Stiglitz show that perfectly informationally efficient markets are inconsistent with costly information acquisition: if prices revealed all information without reward, agents would not pay to produce that information [39]. Conversely, the Milgrom–Stokey no-trade theorem shows that, under common priors, common knowledge of rationality, risk aversion, and an initially Pareto-optimal allocation, private information alone cannot induce mutually agreeable non-null trade [40]. These are equilibrium boundaries, not substitutes for the FTAP.

a. When the Trader Changes the Market The separation between trader and market becomes endogenous when a strategy is large, crowded, or public. Its orders consume liquidity and move quotes; its signals invite imitation or anticipatory trading; and its success changes the distribution on which it was estimated. In that case the deployed price path is better written P^{σ}: it is partly generated by the use of σ itself. The analogy is an invasive classical measurement or feedback experiment, in which the probe perturbs the observed process. In finance the mechanisms are price impact and strategic adaptation, not a physical measurement postulate.

Lo's Adaptive Markets Hypothesis [41] supplies a less idealized interpretation. As capital enters a successful strategy, competition, price impact, and strategic response may erode its edge. The market need not become the tailored countermarket of Theorem 1; that logical construction is fixed from the trader's code and does not model actual price impact. Adaptation merely makes the physical distribution endogenous to widespread strategy use. Nor does impact prove that every large strategy loses; it invalidates naive extrapolation from a backtest that assumes unlimited scale without changing the environment. Feedback may also be amplified when many systems share signals, funding constraints, or risk controls; this is an economic control-and-interaction mechanism, not a consequence of diagonalization or undecidability.

Empirical strategy discovery adds a separate multiple-testing problem. Trying many specifications and reporting the best backtest makes its apparent Sharpe ratio an order statistic rather than an unbiased performance estimate. The Deflated Sharpe Ratio corrects for selection, non-normal returns, and the number of trials [42]. Such corrections do not prove an edge; they make a finite-sample FAPP claim more honestly falsifiable.

X. CONCLUSION

Everything returns to the same pair: a trader and a market. Fix any total computable trader. By simulating that program on the history generated so far, one obtains a fixed total computable countermarket whose

TABLE III. Different meanings of a successful or universal strategy.

Notion	Guarantee	Status
Pathwise universal profit	Positive discounted zero-cost gain on every admissible path	Excluded by classical NA
Relative arbitrage	Outperformance of a specified market or numeraire portfolio	Possible under structural market assumptions
Cover universality	Vanishing per-period log regret to the best constant-rebalanced portfolio	Pathwise but benchmark-relative
FAPP or statistical edge	Positive performance under a specified physical law, data protocol, and horizon	Conditional and empirically falsifiable

next bounded price move has the opposite sign from the trader's position. This is the standard diagonal architecture of universal-inference impossibility: $\forall\sigma\,\exists\mathcal{E}_\sigma$. No physical market has to observe the deployed trader. The always-flat trader earns zero, while every nonzero position incurs a strict loss on its constructed countermarket.

The direct diagonal construction needs no Halting-Problem reduction. The separate halting argument shows that general profit certification and market-rule inference are undecidable even under promises of totality. A passive Martin-Löf-random record blocks unbounded success by effective fair-game capital processes relative to its computable reference measure. No-arbitrage excludes the standard almost-sure zero-cost payoff, and uniform binary averaging fixes expected directional accuracy at 1/2. These are distinct claims with distinct quantifiers.

The Wheel's put-like payoff, Cover's benchmark-relative guarantee, risk premia, informational advantages, and finite-horizon FAPP strategies show what remains possible. A large or public strategy may also change the market on which it was estimated. Accordingly, any defensible claim of repeatable success must state its market restriction, probability law, benchmark, financing and admissibility conditions, horizon, data protocol, and deployment effects. Time reversal is one simple robustness diagnostic among many, not a universal test. The paper's central conclusion is therefore conditional: useful strategies may exist, but no insight-free, zero-cost rule guarantees strict profit against every admissible market.

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OpenAI Codex (GPT-5) was used to assist with literature organization, LaTeX editing, and checks of derivations. The author specified the scientific direction and prompts; model output retained in the manuscript was checked against the cited sources and explicit calculations. The author accepts full responsibility for the final text.

DATA AVAILABILITY

This is a purely mathematical work, and no data or software were created or analyzed in this study.

[1] M. E. Gold, Language identification in the limit, Information and Control **10**, 447 (1967).

[2] L. M. Adleman and M. Blum, Inductive inference and unsolvability, The Journal of Symbolic Logic **56**, 891 (1991).

[3] D. Angluin and C. H. Smith, Inductive inference: Theory and methods, ACM Computing Surveys **15**, 237 (1983).

[4] A. Svozil and K. Svozil, Induction and physical theory formation as well as universal computation by machine learning (CRC Press, Taylor & Francis Group, LLC, Boca Raton, FL, 2021) pp. 170–180, arXiv:1609.03862.

[5] C. D. Moore, Unpredictability and undecidability in dynamical systems, Physical Review Letters **64**, 2354 (1990), cf. Ch. Bennett, *Nature*, **346**, 606 (1990).

[6] L. Bachelier, Théorie de la spéculation, Annales Scientifiques de l'École Normale Supérieure **17**, 21 (1900).

[7] P. A. Samuelson, Proof that properly anticipated prices fluctuate randomly, Industrial Management Review **6**, 41 (1965).

[8] E. F. Fama, Efficient capital markets: A review of theory and empirical work, The Journal of Finance **25**, 383 (1970).

[9] A. M. Turing, On computable numbers, with an application to the Entscheidungsproblem, Proceedings of the London Mathematical Society 2, **42**, 230 (1937).

[10] L. Blum and M. Blum, Toward a mathematical theory of inductive inference, Information and Control **28**, 125 (1975).

[11] D. A. Martin, Borel determinacy, Annals of Mathematics **102**, 363 (1975).

[12] J. Ville, *Étude critique de la notion de collectif*, Thèses de l'entre-deux-guerres No. 218 (Gauthier-Villars, Paris, 1939) pp. 1–124.

[13] P. Martin-Löf, The definition of random sequences, Information and Control **9**, 602 (1966).

[14] C. P. Schnorr, *Zufälligkeit und Wahrscheinlichkeit*, Springer Lecture Notes in Mathematics, Vol. 218 (Springer, Berlin, 1971).

[15] A. Nies, *Computability and Randomness*, Oxford Logic Guides, Vol. 51 (Oxford University Press, Oxford, 2009).

[16] R. G. Downey and D. R. Hirschfeldt, *Algorithmic Randomness and Complexity*, Theory and Applications of Computability (Springer, Berlin, 2010).

[17] C. S. Calude, *Information and Randomness—An Algorithmic Perspective*, 2nd ed. (Springer, Berlin, 2002).

[18] G. Shafer and V. Vovk, *Probability and Finance: It's Only a Game!* (John Wiley & Sons, New York, 2001).

[19] H. G. Rice, Classes of recursively enumerable sets and their decision problems, Transactions of the American Mathematical Society **74**, 358 (1953).

[20] C. Kardaras, Market viability via absence of arbitrage of the first kind, Finance and Stochastics **16**, 651 (2012).

[21] I. Karatzas and C. Kardaras, The numéraire portfolio in semimartingale financial models, Finance and Stochastics **11**, 447 (2007).

[22] R. Fernholz, I. Karatzas, and C. Kardaras, Diversity and relative arbitrage in equity markets, Finance and Stochastics **9**, 1 (2005).

[23] E. Platen and D. Heath, *A Benchmark Approach to Quantitative Finance*, Springer Finance (Springer, Berlin, 2010).

[24] J. M. Harrison and S. R. Pliska, Martingales and stochastic integrals in the theory of continuous trading, Stochastic Processes and their Applications **11**, 215 (1981).

[25] F. Delbaen and W. Schachermayer, A general version of the fundamental theorem of asset pricing, Mathematische Annalen **300**, 463 (1994).

[26] F. Delbaen and W. Schachermayer, The fundamental theorem of asset pricing for unbounded stochastic processes, Mathematische Annalen **312**, 215 (1998).

[27] J. L. Doob, *Stochastic Processes* (John Wiley & Sons, New York, 1953).

[28] D. H. Wolpert and W. G. Macready, No free lunch theorems for optimization, IEEE Transactions on Evolutionary Computation **1**, 67 (1997).

[29] C. Schumacher, M. D. Vose, and L. D. Whitley, The no free lunch and problem description length, in *Proceedings of the Genetic and Evolutionary Computation Conference* (Morgan Kaufmann, San Francisco, 2001) pp. 565–570.

[30] U. G. Haussmann and É. Pardoux, Time reversal of diffusions, The Annals of Probability **14**, 1188 (1986).

[31] K. J. Arrow, *Social Choice and Individual Values*, 3rd ed. (Yale University Press, New Haven, CT, 2012) foreword to the Third Edition by Eric S. Maskin.

[32] N. S. Yanofsky, A universal approach to self-referential paradoxes, incompleteness and fixed points, Bulletin of Symbolic Logic **9**, 362 (2003), arXiv:math/0305282.

[33] G. Bakshi and N. Kapadia, Delta-hedged gains and the negative market volatility risk premium, The Review of Financial Studies **16**, 527 (2003).

[34] P. Carr and L. Wu, Variance risk premiums, The Review of Financial Studies **22**, 1311 (2009).

[35] O. Peters, The ergodicity problem in economics, Nature Physics **15**, 1216 (2019).

[36] T. M. Cover, Universal portfolios, Mathematical Finance **1**, 1 (1991).

[37] J. S. Bell, Against 'measurement', Physics World **3**, 33 (1990).

[38] J. L. Kelly, A new interpretation of information rate, The Bell System Technical Journal **35**, 917 (1956).

[39] S. J. Grossman and J. E. Stiglitz, On the impossibility of informationally efficient markets, The American Economic Review **70**, 393 (1980).

[40] P. Milgrom and N. Stokey, Information, trade and common knowledge, Journal of Economic Theory **26**, 17 (1982).

[41] A. W. Lo, The adaptive markets hypothesis: Market efficiency from an evolutionary perspective, The Journal of Portfolio Management **30**, 15 (2004).

[42] D. H. Bailey and M. López de Prado, The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality, The Journal of Portfolio Management **40**, 94 (2014).